

Notes

Kinds of Function

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(i) One-one function: Let $f: A \rightarrow B$ be a function and it is called one-one if $x \neq y \Rightarrow f(x) \neq f(y)$ for $x, y \in A$ or if $f(x) = f(y) \Rightarrow x = y$.

Ex: Let $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = ax + b$, $a \neq 0$

$$\text{let } f(x), f(y)$$

$$\Rightarrow ax + b = ay + b$$

$$\Rightarrow ax = ay$$

$$\Rightarrow x = y \Rightarrow \text{one-one.} //$$

Note: If f is not one-one then it is called many-to-one.

(ii) Onto function: A function $f: A \rightarrow B$ is called onto if $\text{Range}(f) = B$. (i.e., range = co-domain)

Note: If f is not onto then it is called into.

(iii) Bijective: Let $f: A \rightarrow B$ be a function and it is called bijective if f is one-one and onto.

Ex: The function $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2x - 3$ is bijective.

Solⁿ: For one-one

$$\text{let } f(x) = f(y) \Rightarrow 2x - 3 = 2y - 3 \Rightarrow 2x = 2y$$

$$\therefore x = y \Rightarrow f \text{ is one-one.}$$

Notes

For onto

Suppose that $\exists y \in \text{Range}(f)$ such that $y = 2x - 3$

$$\Rightarrow x = \frac{y+3}{2} \in \mathbb{R}$$

$$\Rightarrow \text{Range of } f = \mathbb{R}$$

It means f is onto.

Hence, f is bijective function.

[But $f: \mathbb{N} \rightarrow \mathbb{N}$, $f(x) = 2x - 3$ is not bijective.]

Que: Show that the mapping $f: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ defined by $f(x) = x^2$ for all $x \in \mathbb{Z}^+$ is one-one but not onto.

Sol? For one-one

$$\text{Let } f(x) = f(y)$$

$$\Rightarrow x^2 = y^2$$

$$\Rightarrow x = y$$

So f is one-one.

For onto

$$\text{If } z \in \text{Range } f$$

$$z = x^2 \Rightarrow x = \sqrt{z} \in \mathbb{Z}^+$$

But for $z = 2$ $\nexists$ s.t. $f(x) = 2$

$\Rightarrow f(x)$ is not onto.

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Inverse function

If $f: X \rightarrow Y$ be a bijective function then there always exist pre-image $f^{-1}(y)$ of each element y of Y and this will be a unique element of X .

Ex: let $f: R \rightarrow R$ such that $f(x) = 2x - 3$.

We know that it is one-one and onto function.
So f^{-1} exist.

$$\text{So, } y = f(x) \Rightarrow x = f^{-1}(y)$$

$$\text{Now, let } y = 2x - 3$$

$$\Rightarrow 2x = y + 3$$

$$\Rightarrow x = \frac{y + 3}{2}$$

$$\Rightarrow f^{-1}(y) = \frac{y + 3}{2} //$$

Composite functions

If $f: A \rightarrow B$ and $g: B \rightarrow C$ be two functions then the function $g \circ f: A \rightarrow C$, where

$$g \circ f(x) = g[f(x)], \forall x \in A \text{ is the composite function of } f \text{ and } g.$$

Ex: If $f: R \rightarrow R$ such that $f(x) = x^2 + 1$ and $g: R \rightarrow R$ such that $g(x) = 2x + 1$ are two functions then $g \circ f: R \rightarrow R$,

$$g \circ f(x) = g[f(x)] = g(x^2 + 1) = 2(x^2 + 1) + 1 = 2x^2 + 3$$

Notes

Que: let $A = \{1, 2, 3\}$. Define $f: A \rightarrow A$ such that $f = \{(1, 2), (2, 1), (3, 3)\}$. Find f^{-1} , f^2 , f^3 .

Hint: $f^{-1} = \{(2, 1), (1, 2), (3, 3)\}$

$f^2 = f \circ f$ and $f^3 = f \circ f \circ f$.

Floor and Ceiling Functions

Let x be any real number. Then x lies between two integers called the floor and the ceiling of x . Specifically,

$\lfloor x \rfloor$, called the floor of x , denotes the greatest integer that does not exceed x .

$\lceil x \rceil$, called the ceiling of x , denotes the least integer that is not less than x .

If x itself is an integer, then $\lfloor x \rfloor = \lceil x \rceil$; otherwise $\lfloor x \rfloor + 1 = \lceil x \rceil$. For example,

$$\lfloor -4 \rfloor = -4, \lfloor 3.14 \rfloor = 3, \lceil \sqrt{5} \rceil = 2, \lfloor -8.5 \rfloor = -9, \lceil 7 \rceil = 7$$

$$\lceil 3.14 \rceil = 4, \lceil \sqrt{5} \rceil = 3, \lfloor -8.5 \rfloor = -8, \lceil 7 \rceil = 7, \lfloor -4 \rfloor = -4$$

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Que: Find the domain of real value function $f(x) = \sqrt{81-x^2}$.

Sol:

We know that $\sqrt{81-x^2} \in \mathbb{R}$.

where $\mathbb{R}$ is the set of real numbers.

$$\therefore 81-x^2 \geq 0$$

$$\Rightarrow x^2 \leq 81$$

$$\Rightarrow |x| \leq 9$$

$$\Rightarrow -9 \leq x \leq 9$$

Hence the domain of f is $[-9, 9]$.

Que: Show that the functions $f(x) = x^4$ and $g(x) = x^{1/4}$ for $x \in \mathbb{R}$ are inverses of one another.

Sol: Let us show that f and g are inverse functions of each other by using property that, two functions f and g are inverses of each other if $f \circ g(x) = I_x$ and $g \circ f(x) = I_x$.

$$\text{Now, } f \circ g(x) = f[g(x)]$$

$$= f(x^{1/4})$$

$$= (x^{1/4})^4 = x = I_x \text{ and}$$

$$g \circ f(x) = g[f(x)]$$

$$= g(x^4)$$

$$= (x^4)^{1/4} = x = I_x$$

Hence $f = g^{-1}$ and $g = f^{-1}$ //

Notes

Que: let $f: A \rightarrow B$ and $g: B \rightarrow A$. Verify that $g \circ f = I_A$.

(a) $A = B = \mathbb{R}$: $f(a) = \frac{a+1}{2}$, $g(b) = 2b-1$

(b) $A = \{x \mid x \text{ is real and } x \geq 0\}$; $B = \{y \mid y \text{ is real and } y \geq -1\}$; $f(a) = a^2-1$, $g(b) = \sqrt{b+1}$.

Solⁿ: (a) Similar as above questions

$$g \circ f(a) = g\left(\frac{a+1}{2}\right) = 2\left(\frac{a+1}{2}\right) - 1 = a$$

$$\Rightarrow g \circ f = I_A //$$

(b) $g \circ f(a) = g(a^2-1) = \sqrt{a^2-1+1} = |a| = a$.

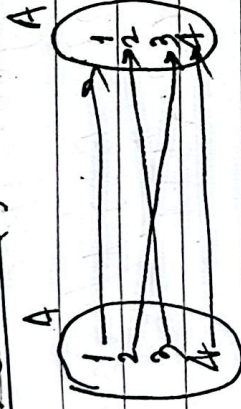
Since $a \geq 0$.

$$\Rightarrow g \circ f = I_A //$$

Notes

Permutation Functions

A bijection from a set A to itself is called a permutation of A .



$$f: \{(1,1), (2,3), (3,2), (4,4)\}$$

$$g: \{(1,1), (2,2), (3,3), (4,4)\}$$

How many bijections can be formed that is the permutation function. Here $4! = 24$ bijective functions can be formed.

Let $A = \{x_1, x_2, x_3, \dots, x_n\}$ be a finite set and P is a bijection on A . That can be represented as

$$P(x_1) \quad P(x_2) \quad P(x_3) \quad \dots \quad P(x_n)$$

Thus, if P is a permutation of finite set $A = \{x_1, x_2, \dots, x_n\}$, then the sequence $P(x_1), P(x_2), \dots, P(x_n)$ is just a rearrangement of the elements of A and so corresponds exactly to a permutation of A .

Ex: Let $A = \{1, 2, 3\}$. Then all the permutations of A are

$$P_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, P_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, P_3 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

$$P_4 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, P_5 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, P_6 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$$

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Inverse of $P.F. (P^{-1})$

Let $P: \begin{bmatrix} a & b & c & d & e \\ c & e & d & a & b \end{bmatrix}$ Hint: i) image row $\rightarrow$ top row
ii) then rearrange of top row.

$$P^{-1} = \begin{bmatrix} c & e & d & a & b \\ a & b & c & d & e \end{bmatrix} \Rightarrow \begin{bmatrix} a & b & c & d & e \\ d & e & a & c & b \end{bmatrix}$$

Composition of P.F.

$$P_1 = \begin{pmatrix} 5 & 10 & 20 & 6 & 4 \\ 10 & 6 & 4 & 5 & 20 \end{pmatrix}$$

$$P_2 = \begin{pmatrix} 5 & 10 & 20 & 6 & 4 \\ 6 & 5 & 10 & 20 & 4 \end{pmatrix}$$

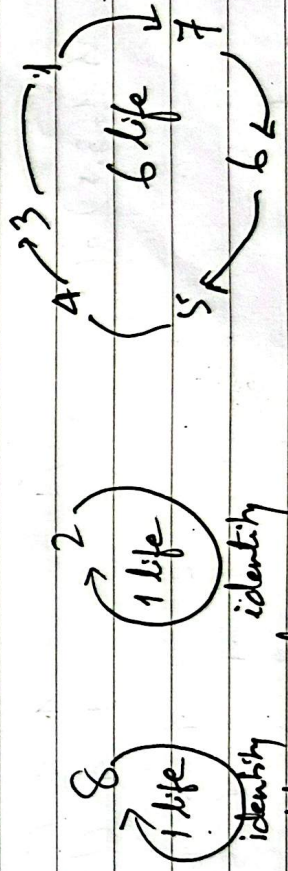
$$\therefore P_1 \circ P_2 = \begin{pmatrix} 5 & 10 & 20 & 6 & 4 \\ 5 & 10 & 6 & 4 & 20 \end{pmatrix}$$

$$[P_1 \circ P_2(5) = P_1[P_2(5)]] = P_1(6) = 5$$

The composition of two permutations is another permutation, usually referred to as the product of these permutations.

Cyclic Permutation

$$P = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 7 & 2 & 1 & 3 & 4 & 5 & 6 & 8 \end{pmatrix}$$



3 possible cycles

length = 1

length = 1

length = 6

Notes

Ex: Find all possible cycles.

$$P = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 7 & 4 & 6 & 1 & 5 & 3 & 2 \end{pmatrix}$$

Sol: $(1, 7, 2, 4), (3, 6), (5)$

Ex: $A = \{a, b, c, d, e, f, g, h\}$

Find product of cycles $\{e, f, c, d, a\}$ and $\{a, b, h, f\}$

Sol: $(ef, c, d, a) = \begin{pmatrix} a & b & c & d & e & f & g & h \\ e & b & d & a & f & c & g & h \end{pmatrix}$

$$(a, b, h, g) = \begin{pmatrix} a & b & c & d & e & f & g & h \\ b & h & c & d & e & f & a & g \end{pmatrix}$$

Now,

$$(e, f, c, d, a) \circ (a, b, h, g) = \begin{pmatrix} a & b & c & d & e & f & g & h \\ e & b & d & a & f & c & g & h \end{pmatrix} \circ \begin{pmatrix} a & b & c & d & e & f & g & h \\ b & h & c & d & e & f & a & g \end{pmatrix}$$

$$= \begin{pmatrix} a & b & e & d & e & f & g & h \\ b & h & d & a & f & c & e & g \end{pmatrix} //$$

- The product of two cycles need not be a cycle.

Example: Let $A = \{1, 2, 3, 4, 5, 6\}$. Compute $(4, 1, 3, 5) \circ (5, 6, 3)$

and $(5, 6, 3) \circ (4, 1, 3, 5)$.

Sol:

$$(4, 1, 3, 5) \circ (5, 6, 3) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 2 & 5 & 1 & 4 & 6 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 5 & 4 & 6 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 2 & 4 & 1 & 6 & 5 \end{pmatrix}$$

and

$$(5, 6, 3) \circ (4, 1, 3, 5) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 2 & 6 & 1 & 4 & 3 \end{pmatrix}$$

Notes

observe that $(4, 1, 3, 5) \circ (5, 6, 3) \neq (5, 6, 3) \circ (4, 1, 3, 5)$ and that neither product is a cycle.

- Two cycles of a set A are said to be disjoint if no element of A appears in both cycles.
For example: let $A = \{1, 2, 3, 4, 5, 6\}$. Then the cycles $(1, 2, 5)$ and $(3, 4, 6)$ are disjoint, whereas the cycles $(1, 2, 5)$ and $(2, 4, 6)$ are not.

Theorem: A permutation of a finite set that is not the identity or a cycle can be written as a product of disjoint cycles of length ≥ 2 .
(Proof not necessary.)

Example: Write the permutation

$$P = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 4 & 6 & 5 & 2 & 1 & 8 & 7 \end{pmatrix}$$

on the set $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$ as a product of disjoint cycles.

Solution: We start with 1 and find that $P(1)=3, P(3)=6$, and $P(6)=1$. So we have the cycle $(1, 3, 6)$. Next we choose the first element of A that has not appeared in a previous cycle. We choose 2, and we have $P(2)=4, P(4)=5$, and $P(5)=2$, so we obtain the cycle $(2, 4, 5)$. We now choose 7, the first element of A that has not appeared in a previous cycle. Since $P(7)=8$ and $P(8)=7$, we obtain the cycle $(7, 8)$. We can then write P as a product of disjoint cycles as

$$P = (7, 8) \circ (2, 4, 5) \circ (1, 3, 6).$$

(The product is unique except for the order of the cycles.)

Even and Odd Permutations

A permutation is said to be an even permutation if it can be expressed as a product of an even number of transpositions, otherwise it is said to be an odd permutation.

Transposition: Any cycle of length two is called a transposition.

Eg: The cycle $(2, 3)$ is a transposition.

A cycle can be decomposed into transpositions.

e.g. let $(2, 3, 4)$ be a cycle. It can be decomposed into $(2, 3) \cdot (2, 4) \cdot (2, 3) \cdot (2, 4)$

Example: Examine whether the following permutation is even or odd.

$$P = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 5 & 4 & 3 & 6 & 1 & 7 & 9 & 8 \end{pmatrix}$$

Sol: let us denote the given permutation by P . Expressing P as a product of disjoint cycles, we can write

$$P = (1 \ 2 \ 5 \ 6) (3 \ 4) (7) (8, 9)$$

Omitting cycle of length 1 because it represents identity permutation.

Now, expressing the cycle $(1, 2 \ 5 \ 6)$ as a product of transpositions, we can write

$$(1 \ 2 \ 5 \ 6) = (1 \ 2) (1 \ 5) (1 \ 6)$$

$$\therefore P = (1 \ 2) (1 \ 5) (1 \ 6) (3 \ 4) (8 \ 9)$$

Notes

Thus P has been expressed as a product of transpositions. The number of transpositions in this product is 5 which is odd.
Hence P is an odd permutation.

Remarks:

1. A permutation cannot be both even and odd.
2. A cycle of length n can be expressed as the product of $n-1$ transpositions.
3. Identity permutation is always an even permutation.
4. The product of two odd permutations is an even permutation.
5. The product of two even permutations is an even permutation.
6. The product of an even permutation and an odd permutation is an odd permutation.

H/w: Let $P_1 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 7 & 3 & 2 & 1 & 4 & 5 & 6 \end{pmatrix}$ and

$$P_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 6 & 3 & 2 & 1 & 5 & 4 & 7 \end{pmatrix}.$$

(a) Compute $P_1 \circ P_2$.

(b) Compute P_1^{-1} .

(c) Is P_1 an even or odd permutation? Explain.